

1. Introduction

Payroll management is an important administrative task in any university. Manual payroll handling is often time-consuming and error-prone. To address this issue, a web-based University Payroll Management System was developed.

This laboratory project mainly focuses on the frontend design and implementation of the system. The frontend provides an interactive interface that allows administrators to manage employees, process salaries, and view payroll-related information easily.

2. Objectives

The objectives of this project are:

- To design a simple and user-friendly frontend interface
- To manage employee information through web forms
- To visualize payroll data using tables and charts
- To improve usability through dashboard-based navigation

3. System Overview

The system follows a client–server architecture where the frontend acts as the main interaction layer. All operations such as adding employees, paying salaries, and viewing reports are performed through the web interface.

The frontend is designed with a dashboard layout and sidebar navigation to ensure ease of use and quick access to different features.

4. Frontend Features

The main frontend features of the system include:

- Employee management through form-based input
- Display of employee records in tabular format
- Monthly salary payment interface with clear feedback
- Payroll summary visualization with basic filters
- Salary receipt generation with structured layout

5. User Interface Description

The user interface is designed to be clean and visually comfortable. A sidebar menu is used for navigation, while cards, tables, and charts are used to present information clearly. Feedback messages are shown to inform the user about successful or invalid actions.

6. Software Interface Pages

1. Home Page: Home page of the software has employee count and their distribution by various category.

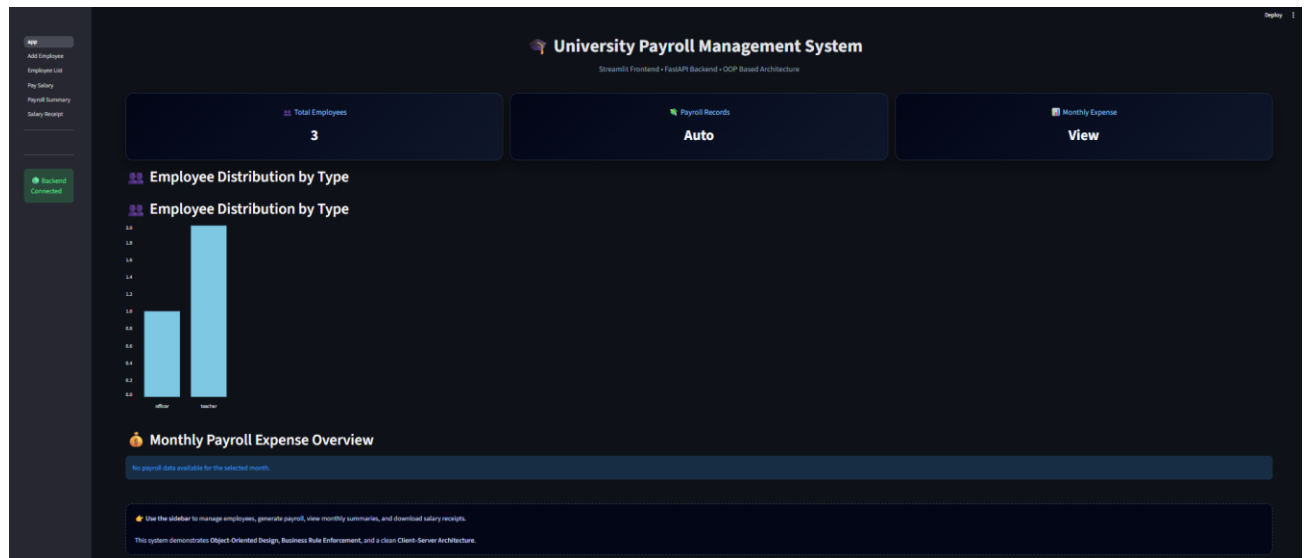


Figure 01: Home Page

2. Add Employee: Add employee page contains a form regarding the information of the employee. Which includes Employee type, Full Name, Employee Age, Email address, Phone Number, Designation, Faculty, Department and Basic Salary.

The screenshot shows the 'Add New Employee' form. The left sidebar contains links for 'app', 'Add Employee', 'Employee List', 'Pay Salary', 'Payroll Summary', and 'Salary Receipt'. The main form area is titled '+ Add New Employee' and includes the instruction 'Fill in the employee details below'. The form fields are arranged in two columns: Employee Type (dropdown), Designation (text), Full Name (text), Faculty (text), Age (text with min/max), Department (text), Email (text), Basic Salary (text with min/max), and Phone Number (text). An 'Add Employee' button is located at the bottom left of the form.

Figure 02: Add New Employee Form

3. Employee list: Employee List page shows the list of the current employees.

The screenshot shows a web application interface with a dark theme. On the left is a sidebar menu with options: 'app', 'Add Employee', 'Employee List' (highlighted), 'Pay Salary', 'Payroll Summary', and 'Salary Receipt'. The main content area is titled 'Employee List' with a purple icon of two people. Below the title is a table with 8 columns: 'id', 'name', 'emp_type', 'age', 'designation', 'faculty', 'department', and 'basic_salary'. The table contains three rows of employee data. Below the table is a 'Delete Employee' section with a red prohibition icon, a text input field for 'Enter Employee ID' containing the value '1', a checkbox labeled 'I confirm Deletion of this employee', and a 'Delete Employee' button.

id	name	emp_type	age	designation	faculty	department	basic_salary
1	Rayhan	officer	21	Chief Engineer	Engineering	Software Engineering	275999
2	Marwan	teacher	20	Professor	Engineering	CSE	350000
5	Emran	teacher	18	Associate Professor	BBA	Marketing	230000

Figure 03: Employee List And Employee Deletion Page

4. Pay Monthly Salary: This page gives admin to pay the salary of all the employees for a particular year and month. Salary of a single month can not be given twice.

The screenshot shows the 'Pay Monthly Salary' page in the same web application. The sidebar menu is identical, with 'Pay Salary' highlighted. The main content area is titled 'Pay Monthly Salary' with a green icon of a dollar sign and a calendar. Below the title is the subtitle 'Generate salary for all employees for a selected month'. There are two dropdown menus for 'Year' (set to 2025) and 'Month' (set to August). Below these is a green box with 'Important Rules' containing a bullet point: 'A month's salary cannot be paid twice for the same employee.' There is a 'Pay Salary for All Employees' button. A green success message states 'Salary processed successfully!'. Below this is a 'Result' section showing a JSON object:

```
{  "created": [    0 : 1    1 : 2    2 : 5  ],  "errors": {}}
```

Figure 04: Pay Monthly Salary of the Employees

5. Payroll Summary:

Monthly salary expenditure report

Year: 2025 Month: March

Optional Filters

Faculty (optional) Department (optional) Designation (optional)

Generate Summary

Summary generated successfully

Total Salary Paid

1050318.78

```
{
  "year": 2025,
  "month": 3,
  "total_paid": 1050318.78
}
```

Figure 05: Payroll Summary With Filters on Employees

6. Salary Receipt:

Salary Receipt

Employee-wise monthly salary slip

Employee ID: 2 Year: 2025 Month: March

View Salary Receipt

Salary receipt generated

Employee Information

Name: Marwan Department: CSE
Designation: Professor Month: 3/2025
Faculty: Engineering

Salary Breakdown

Allowances

Basic Salary: 350000.0
House Rent Allowance: 70000.0
Travel Allowance: 35000.0
Medical Allowance: 2000.0

Deductions

VAT/TAX: 17500.0
Provident Fund: 10500.0

Net Salary

429000.0

Figure 06: Salary Receipt of the Employee

7. Testing and Validation

Frontend testing was performed by navigating all pages, submitting forms, and verifying proper display of tables, charts, and feedback messages. All interfaces worked as expected.

8. Conclusion

This lab project demonstrates the frontend implementation of a University Payroll Management System. The system provides a simple and effective web interface for payroll operations, fulfilling the requirements of a software engineering laboratory assignment.